

JUNDA ZHAO

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Research Interests

Reliable testing and verification for AI coding agents in production: LLM-based test generation from recovered specifications, and the evaluation of AI-generated tests. **Seeking a postdoctoral position starting September 2027.**

Education

Doctor of Philosophy (Ph.D. Candidate), Mechanical and Industrial Engineering 09/2022 to present
University of Toronto, St. George Campus, Toronto, Canada

- Advisor: Prof. Eldan Cohen, Optimization and Machine Learning (OptiMaL) Lab; close collaboration with Prof. Shurui Zhou (Electrical and Computer Engineering)
- Dissertation (in progress): *Accurate Specification Recovery for Effective Test Generation*
- Passed the doctoral qualifying examination; expected completion in 2027

Honours Bachelor of Science, Computer Science 09/2016 to 11/2021
University of Toronto, St. George Campus, Toronto, Canada

Selected coursework

- Graduate: CSC2511 Natural Language Computing (A+); MAT1510 Deep Learning: Theory and Data Science (A+); MIE1513 Decision Support Systems (A+)
- Undergraduate: CSC412 Probabilistic Learning and Reasoning (A+/94); CSC413 Neural Networks and Deep Learning (A+/99); CSC420 Introduction to Image Understanding (A+/96)

Honours and Awards

ACM SIGSOFT Distinguished Paper Award, ISSTA 2026 2026
For “Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness? (Replicability Study)”

Publications

- [1] **Junda Zhao**, Shurui Zhou, Eldan Cohen. “Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness? (Replicability Study).” *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026)*. **ACM SIGSOFT Distinguished Paper Award**. [arXiv](#)
- [2] **Junda Zhao**, Shurui Zhou, Eldan Cohen. “Evaluating and Mitigating the Misguidance Effect of Buggy Code in LLM-Generated Unit Tests.” *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026)*. [arXiv](#)
- [3] **Junda Zhao**, Yuliang Song, Eldan Cohen. “Variational Prefix Tuning for Diverse and Accurate Code Summarization Using Pre-trained Language Models.” *Journal of Systems and Software*, 2025. [Paper](#)
- [4] Faria Khandaker, Arik Senderovich, **Junda Zhao**, Eldan Cohen, Eric Yu, Sebastian Carbajales, Allen Chan. “Transformer Models for Mining Intents and Predicting Activities from Emails in Knowledge-Intensive Processes.” *Engineering Applications of Artificial Intelligence*, 2024. [Paper](#)
- [5] **Junda Zhao**, Weiling Cai, Yizhen Ge, Lijuan Zhang. “A New Fuzzy Belonging-based Multi-view K-means Clustering Algorithm.” *IEEE Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON 2019)*. [Paper](#)
- [6] **Junda Zhao**. “Accurate Specification Recovery for Effective Test Generation.” *SPLASH/ISSTA 2026 Doctoral Symposium*.

Research Experience

Ph.D. Candidate, OptiMaL Lab, Department of Mechanical and Industrial Engineering 09/2022 to present
University of Toronto, supervised by Prof. Eldan Cohen, Toronto, Canada

- **LLM-based unit test generation.** Introduced a metric for the misguidance effect, where buggy code steers LLMs toward tests that validate the bug rather than expose it. Evaluated 11 LLMs on 318 focal methods from Defects4J and proposed specification-based test generation, which reduces misguided tests while substantially increasing effective tests (ISSTA 2026).
- **Evaluation of LLM-generated tests.** Led a large-scale replication of the Inozemtseva and Holmes and the Papadakis et al. studies on more than 100,000 test cases generated by 11 LLMs for Defects4J, showing that coverage and mutation scores are meaningful signals in regression-style settings but not reliable indicators when the code under test may already be buggy (ISSTA 2026, ACM SIGSOFT Distinguished Paper Award).
- **Specification recovery.** Dissertation research on accurate, evidence-driven, and uncertainty-aware recovery of behavioral specifications from imperfect code, so that generated tests detect bugs.
- **Code summarization.** Developed Variational Prefix Tuning, a parameter-efficient method that integrates a conditional variational autoencoder into pre-trained language models of code to generate diverse yet accurate summaries (JSS 2025).

- **Open research artifacts.** Released public replication packages for both ISSTA 2026 papers on GitHub, archived on Zenodo ([10.5281/zenodo.21429528](https://zenodo.org/record/21429528), [10.5281/zenodo.21428153](https://zenodo.org/record/21428153)).

Research Assistant, Dynamic Graphics Project (DGP) Lab

11/2021 to 09/2022

University of Toronto, Toronto, Canada

- Research on geometric stylization: developed a method for texture synthesis and transfer based on a feature-grid neural implicit representation of arbitrary objects, whose 2-D structure allows computer vision and 2-D geometry processing techniques to be applied to 3-D representations.

Undergraduate Projects, Department of Computer Science, University of Toronto

10/2020 to 09/2021

- **Human musculoskeletal system simulation** (CSC494 research project): proposed, built, and simulated a fundamental unit of the musculoskeletal system from mass points, rods, and springs, with machine learning for fast approximation of the system state.
- **Accurate large-scale ferrofluid simulation** (CSC417 team course project): implemented the ferrofluid simulation of Huang et al. and studied the formation of spikes under a magnetic field.

Talks and Posters

Research-track talk and poster, ISSTA 2026, Oakland, CA, USA

10/2026

“Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness? (Replicability Study)”

Research-track talk and poster, ISSTA 2026, Oakland, CA, USA

10/2026

“Evaluating and Mitigating the Misguidance Effect of Buggy Code in LLM-Generated Unit Tests”

Journal-First talk and poster, ASE 2025, Seoul, South Korea

11/2025

“Variational Prefix Tuning for Diverse and Accurate Code Summarization Using Pre-trained Language Models”

Doctoral Symposium talk, SPLASH/ISSTA 2026, Oakland, CA, USA

10/2026

“Accurate Specification Recovery for Effective Test Generation”

Teaching Experience

Teaching Assistant, MIE1517 Introduction to Deep Learning (graduate course)

01/2024 to 04/2024; 09/2024 to present

University of Toronto, Department of Mechanical and Industrial Engineering

- Fall and Winter terms since 09/2024. Lead tutorial sessions, help develop and improve course materials and plans, and grade projects and tests.

Teaching Assistant, MIE370 Introduction to Machine Learning (undergraduate course)

09/2025 to 12/2025

University of Toronto, Department of Mechanical and Industrial Engineering

- Led tutorial sessions, helped prepare teaching materials, and graded projects and tests.

Teaching Assistant, MIE350 Design and Analysis of Information Systems (undergraduate course)

09/2022 to 12/2022

University of Toronto, Department of Mechanical and Industrial Engineering

- Led tutorial sessions, helped prepare teaching materials, and graded projects and tests.

Professional Experience

Junior Identity and Access Management Analyst

09/2019 to 09/2020

University of Toronto, Information Technology Services, Toronto, Canada

- Built and maintained a database system for tracking and analyzing network access and activity information for students, faculty, and staff on the university network.

Skills

- **Programming:** Python, C++ (Eigen, libigl), MATLAB; Java for test generation and evaluation (JUnit, Defects4J)
- **Machine learning:** PyTorch, NumPy, OpenCV; large language models and deep generative models; statistical analysis and modeling
- **Software testing and analysis:** LLM-based test generation, coverage and mutation analysis (JaCoCo, PIT, CodeCover), experiment design
- **Languages:** fluent in English and Mandarin